

Hieu Vu

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SUMMARY

MSc student in Distributed Systems at KTH, seeking internship and thesis roles in platform and ML-systems engineering. Previously spent two years as a Cloud Platform Engineer on the founding team behind [Managed FPT Kubernetes Engine \(M-FKE\)](#), building Kubernetes control-plane components in Go serving 500+ customer clusters across Vietnam and Japan. Currently working on LLM serving, distributed consensus, and multi-agent LLM systems.

EDUCATION

- 2025 – 2027 MSc Software Engineering of Distributed Systems**, KTH Royal Institute of Technology
Courses: Advanced Distributed Systems (A), Data-Intensive Computing, Data Mining, HPC, Network Systems for ML
- 2019 – 2023 BSc Electronics and Telecommunications**, Hanoi University of Science and Technology
GPA 3.58/4.0 | Ranked **1st in graduation thesis defense (9.5/10)**

WORK EXPERIENCE

- Cloud Platform Engineer, Founding Platform Team – FPT Smart Cloud** *Aug 2023 – Aug 2025*
Founding engineer in a 6-person team that built M-FKE, a managed Kubernetes service similar to AWS EKS.
- Built and operated the Kubernetes-on-Kubernetes control plane behind 500+ customer clusters on OpenStack and VMware vSphere: VM provisioning, cluster bootstrap, CIDR and security groups, API server exposure, and Terraform-based hybrid-cloud provisioning for Gardener MCM/CCM.
 - Built the cluster hibernate and wake-up workflow: worker nodes scale to zero while etcd state, PVCs, and Services are kept, and the cluster is restored on demand. Reduced infrastructure cost by 30% for enterprise customers.
 - Wrote a Go/Kubebuilder Operator for PVC backup and cluster-state recovery, with CRDs and reconciliation loops for scheduled snapshots, retention policies, and on-demand restores.
 - Extended the Cloud Controller Manager to provision L4/L7 load balancers for LoadBalancer Services, with Proxy Protocol and multi-zone high availability. Added Cilium/eBPF as a self-service CNI option at cluster creation.
 - Ran quarterly zero-downtime Kubernetes upgrades (up to v1.32) across multi-site environments.
- Backend Developer Intern – Viettel Cyber Security** *Jun 2022 – Nov 2022*
Selected for the VCS Talent Program. Built a Go/PostgreSQL backend for tracking internal company servers.

PROJECTS

- KV-Cache-Aware LLM Serving** *Python, vLLM, LMCache – KTH course project*
- Built the request generator and benchmark runner for a vLLM + LMCache serving stack (team of 4), and measured how local KV-cache size affects time-to-first-token and end-to-end latency.
 - Wrote a pipeline that identifies a request's source document from its cached KV tensors after inference (PCA + KMeans, NMI 1.00).
- Parallel Electromagnetic Wave Simulation (FDTD)** *C, MPI, OpenMP – KTH course project*
- Parallelized an FDTD solver with MPI domain decomposition and halo exchange (team of 3), reaching 14.8x speedup at 92% efficiency on 16 cores of the Dardel supercomputer; compared against OpenMP, hybrid, and GPU-offload versions using perf.
- Deadline-Ordered Consensus on OmniPaxos** *Rust – KTH course project*
- Added Nezha-style deadline ordering to the OmniPaxos key-value store (team of 4). Built the clock model (drift and uncertainty), the deadline-ordered early buffer, and an adaptive deadline based on 95th-percentile delay; benchmarked how clock quality affects latency and fast-path ratio.

TECHNICAL SKILLS

Languages	Go, Python, Rust, C, Bash, Erlang/OTP
Kubernetes & Cloud	Kubernetes (Operators, CRDs, Kubebuilder), Gardener, Cilium/eBPF, Helm, GitOps, Terraform, OpenStack, VMware vSphere, AWS, Docker, Linux
ML Systems & HPC	vLLM, LMCache, MPI, OpenMP, HuggingFace Transformers, scikit-learn, PySpark, Delta Lake
Data & Observability	PostgreSQL, MySQL, MongoDB, Redis, etcd, Prometheus, Grafana, Loki

CERTIFICATIONS AND AWARDS

- Apr 2026 Winner**, SSEN Hackathon 2026 (Ellipsis VC Challenge): built a multi-agent AI app that automates marketing and finance workflows for one-person companies
- Mar 2026 AWS Certified Solutions Architect – Associate (SAA-C03)**